

# OR Gate



## **What is a NAND Gate?**

**An OR gate is a basic logic gate that produces a HIGH (1) output when any of its inputs (or all of them) are HIGH. The output is LOW (0) only when all inputs are LOW (0).**

# Truth Table



| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

# Function of the OR Gate

**An OR gate performs a logical addition (disjunction) operation on its inputs.**

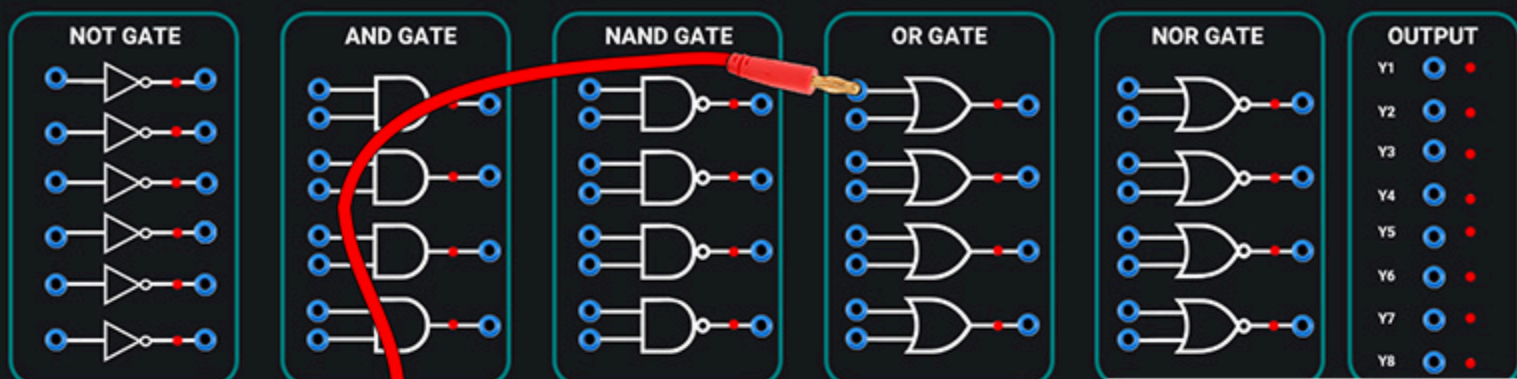
**The main function of an OR gate is to combine two or more input signals and produce an output signal according to this rule:**

**Output = 1 when any input (or all inputs) = 1**

**Output = 0 only when all inputs = 0**



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## Connect the OR Gate

Connect a wire from the gate output to terminal Y1 (or any point from Y1 to Y8) in the OUTPUT section at the top right.

3

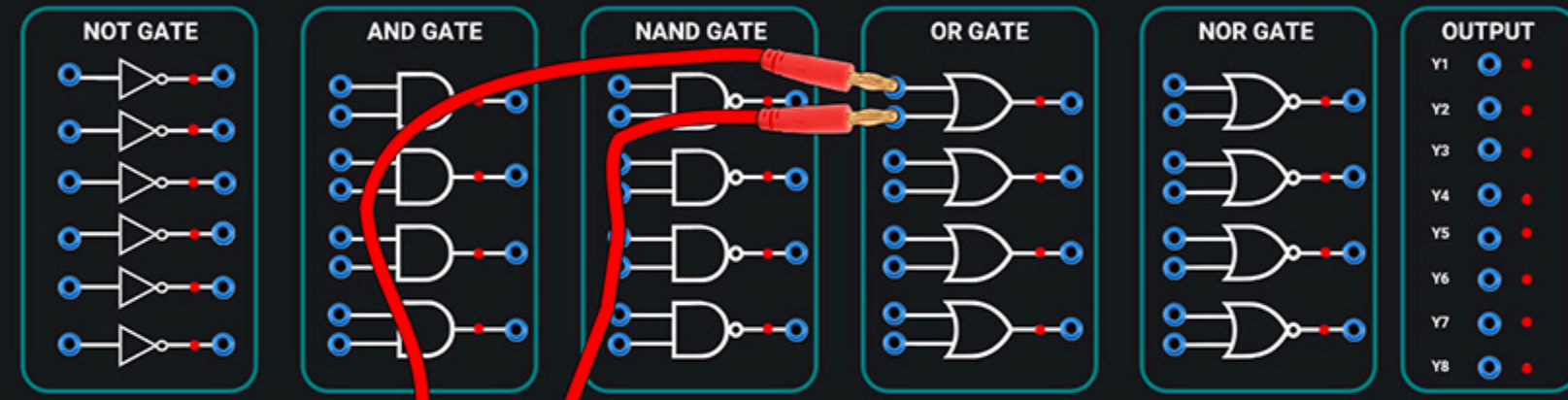
1

Connect a wire from the first input (top blue terminal) to switch A in the INPUT section.

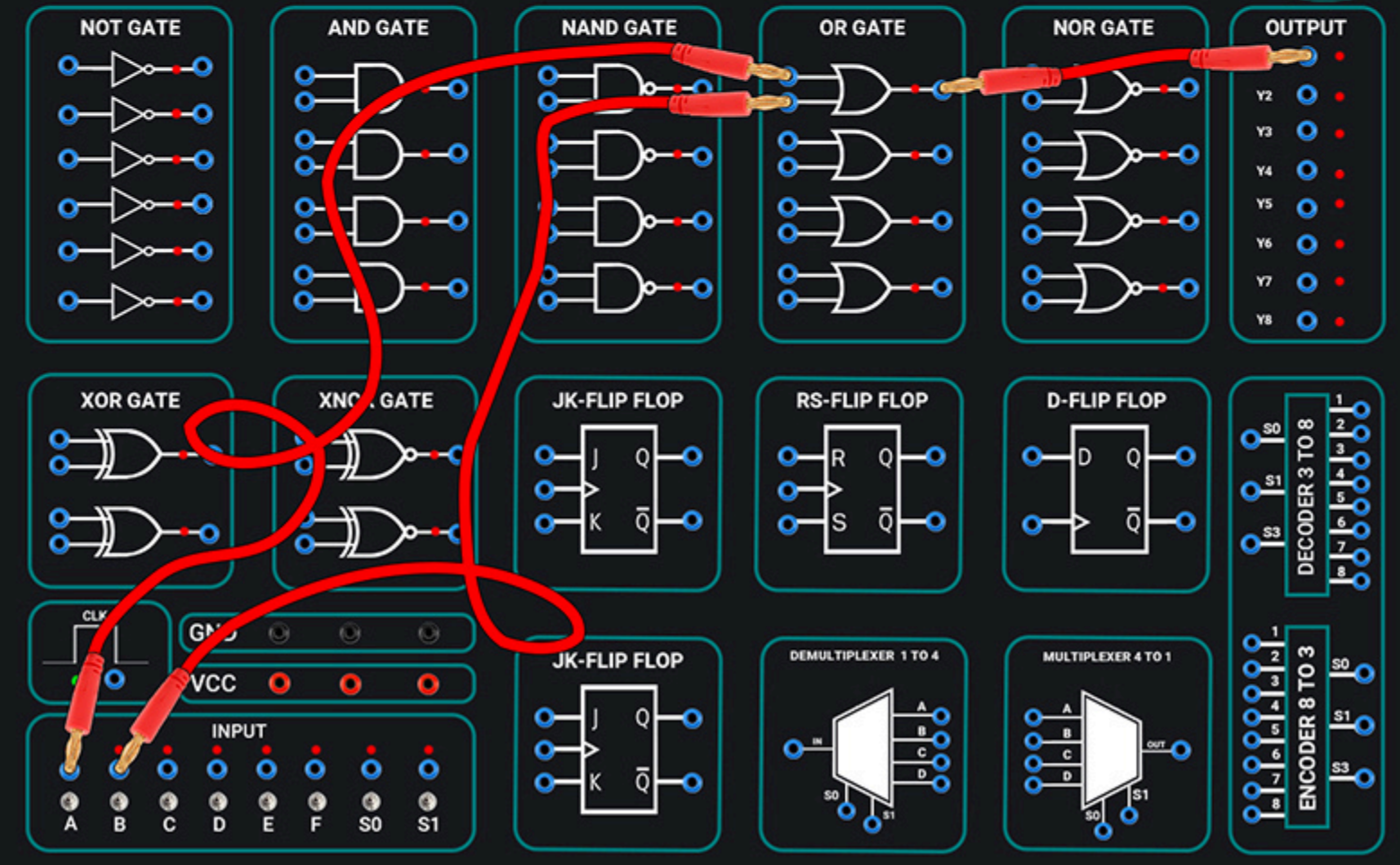
2

Connect a wire from the second input (bottom blue terminal) to switch B in the INPUT section

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